

Individual Reflection

For this assignment, my main contribution was focused on the **network analysis section**, while also supporting the overall report structure, interpretation, and final documentation. I worked on understanding how YouTube users interacted with each other through reply relationships and how this interaction could be represented as a directed network. In this network, each pseudonymous commenter was treated as a node, and each reply relationship was treated as an edge. This helped move the project beyond only analysing comment text, because it allowed us to study how the fan discussion was socially structured.

The most important part of my work was analysing **centrality measures**, including degree, in-degree, PageRank, betweenness, and eigenvector centrality. These measures helped identify which commenters were most influential in different ways. For example, PageRank and in-degree showed which users attracted important reply links, while betweenness showed users who acted as bridges between different parts of the discussion. I also examined clustering coefficients to understand whether the AGP fan network formed tightly connected groups. The results showed that clustering was extremely low, meaning most users replied in simple one-to-one interactions rather than forming dense friendship-style groups. This was an important finding because it showed that the network was large and active, but still structurally sparse.

I also contributed to the **NetworkX visualisations** and community detection interpretation. The PageRank-sized network visualisation helped show that only a small number of users had strong structural importance, while most users were located around the edges of the network. The community detection section was also important because it showed that the discussion was divided into many modular sub-publics. These communities were not dense clusters, but they were still meaningful because they were organised around drivers, teams, and race narratives such as Oscar Piastri, Lando Norris/McLaren, and Max Verstappen.

Working on the whole report helped me understand how different analysis methods connect together. Sentiment analysis explained the emotional tone of the discussion, topic modelling explained what fans were talking about, and network analysis explained how users were connected. One challenge was making sure the report did not just describe figures, but actually interpreted what each result meant for the research question. I learned that strong analysis requires linking statistics, figures, and discussion back to the project aim.

Overall, this assignment improved my understanding of social media and network analytics. I learned how Python tools such as pandas, NetworkX, matplotlib, and community detection methods can be used to analyse real online interaction data. I also learned the importance of presenting results clearly, avoiding oversized tables, and explaining figures in a way that supports the argument of the report. If I improved this work further, I would add more comparison between different videos or time periods to see whether fan communities changed before and after key race events.